

Improving Equilibrium Propagation without Weight Symmetry through Jacobian Homeostasis

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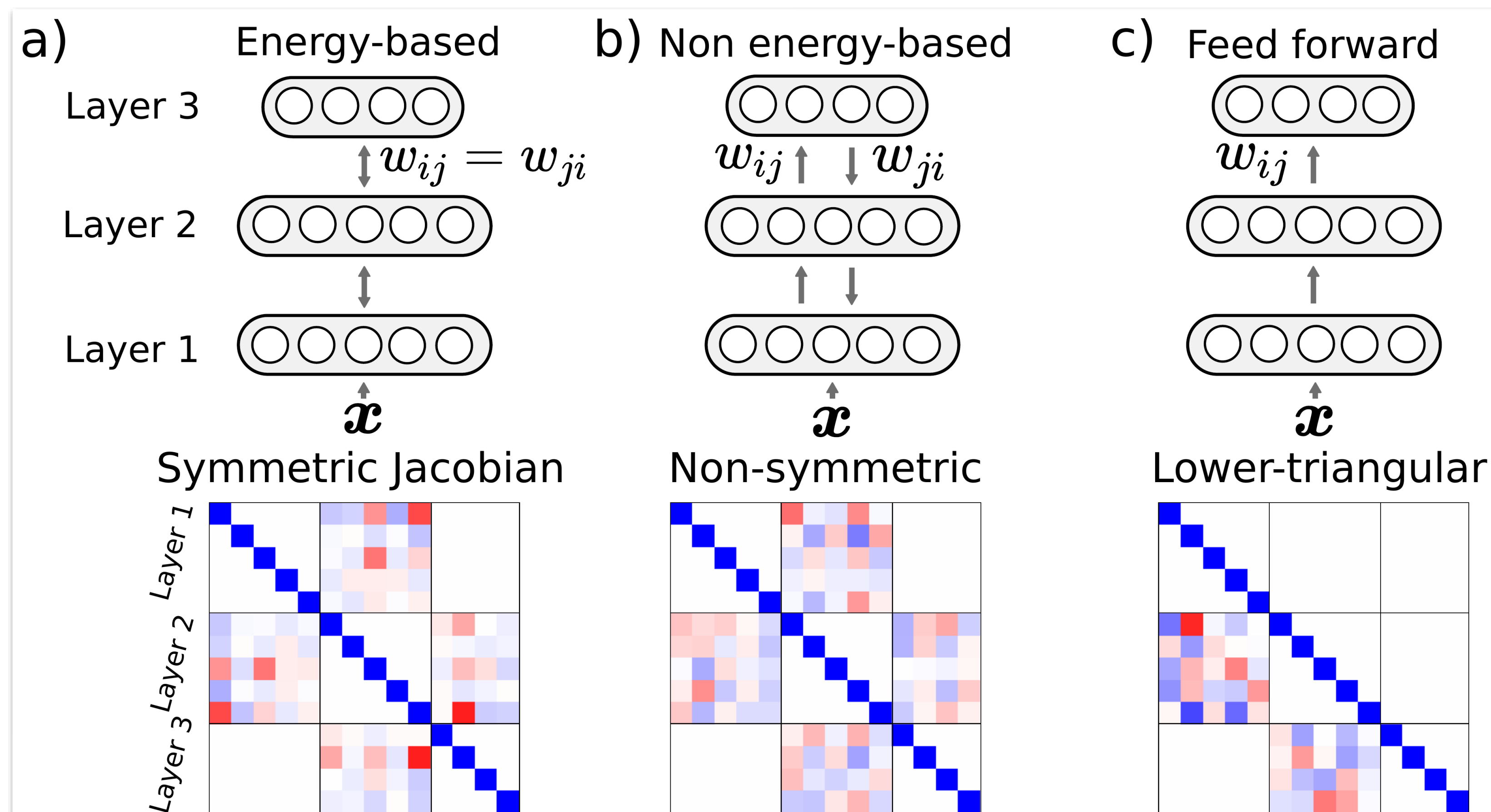
Problem

Equilibrium propagation (EP) computes gradients from a network's own relaxation, but stays unbiased only under two conditions physical hardware cannot meet: perfectly **symmetric weights** and an **infinitesimal nudge**.

Asymmetry's true cost had never been isolated — it hides behind the finite nudge.

Motivation

- Physical substrates (brains, analog chips) could train at a fraction of digital energy, yet backprop needs the exact **transpose** of every weight they cannot provide.
- Generalized (asymmetric) EP had only worked on MNIST and **failed outright on CIFAR-10** — an unexplained scaling failure.



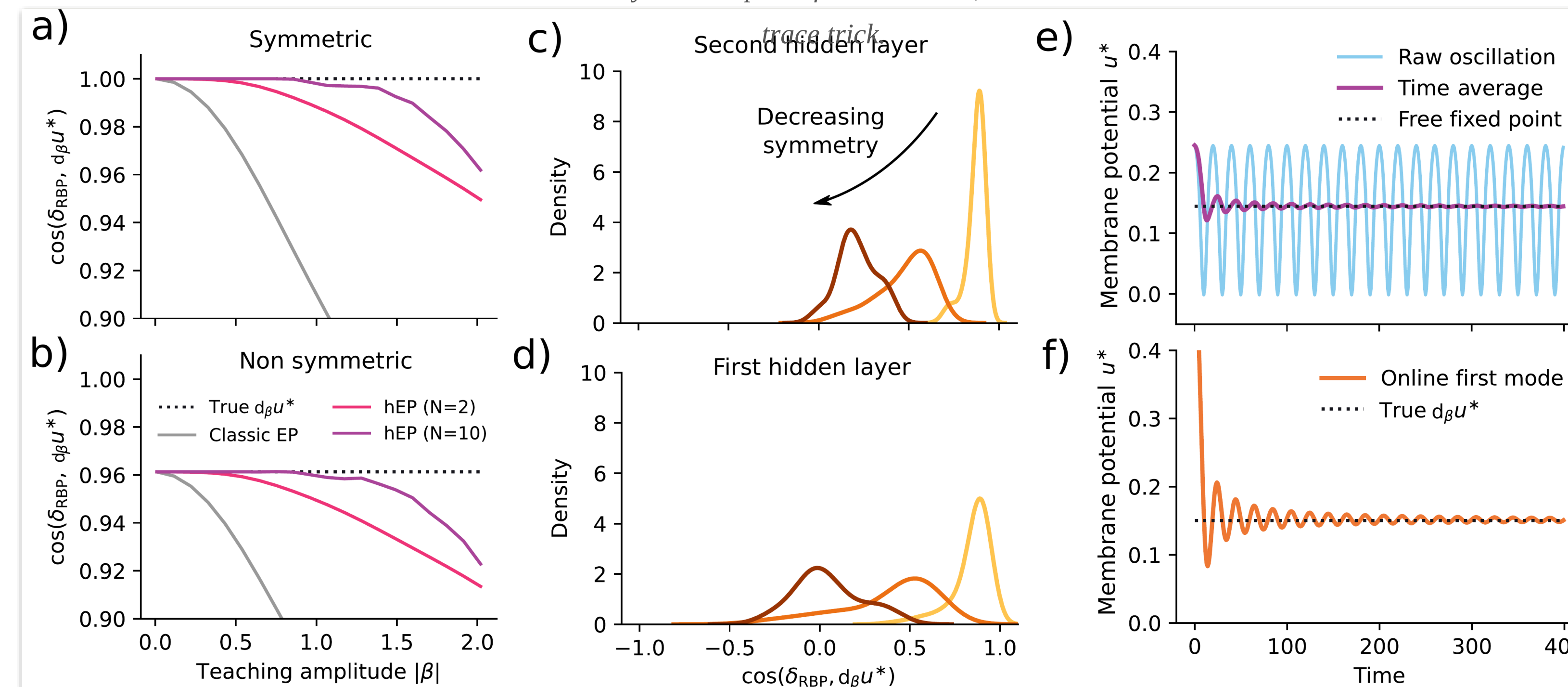
Energy-based nets force weight symmetry (symmetric Jacobian); dropping it breaks symmetry, while feedforward nets stay lower-triangular.

Method

- Generalized holomorphic EP** drives the net with an oscillating teaching signal $\beta(t)$ and reads the exact neuronal error via a **Cauchy integral** — killing finite-nudge bias even for asymmetric Jacobians, in continuous time without separate free/nudged phases.
- A new **homeostatic loss** penalizes the skew-symmetric part A of the free-fixed-point Jacobian, improving functional symmetry without ever tying the weights.

$$\mathcal{L}_{\text{homeo}} = \mathbb{E}_{\varepsilon \sim \mathcal{N}(0, I)} [\|J_F \varepsilon\|^2 - \varepsilon^\top J_F^2 \varepsilon]$$

Penalizes the skew-symmetric part of the Jacobian, estimated with the Hutchinson



The Cauchy integral recovers the exact error at finite nudge, even for asymmetric Jacobians.

Datasets

Bias is first isolated on small MLPs, then hEP is scaled on a recurrent convolutional net with **genuinely asymmetric** feedback weights.

Fashion-MNIST

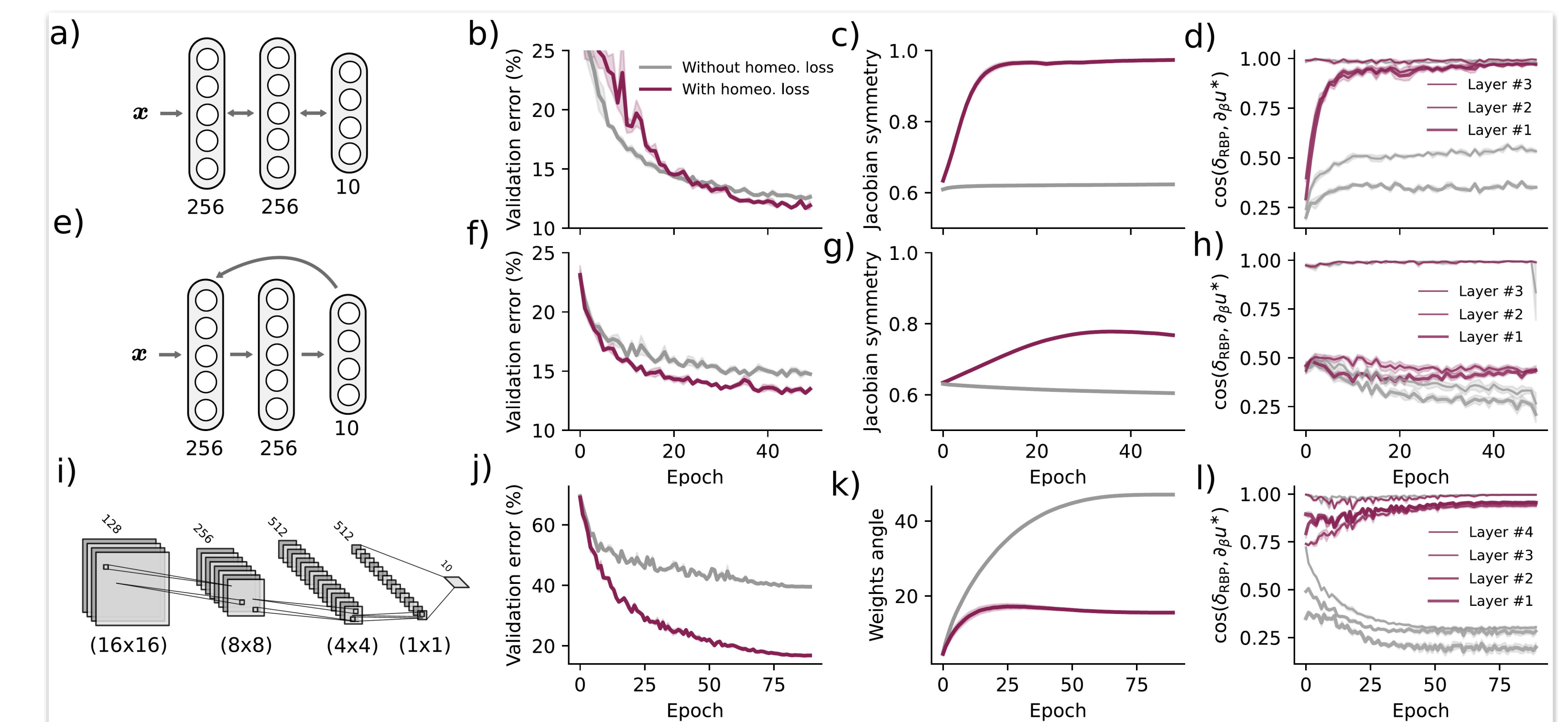
CIFAR-10

CIFAR-100

ImageNet 32x32

Key Results

Asymmetric hEP	C-10	C-100	IN 32 ²
w/o $\mathcal{L}_{\text{homeo}}$	60.4	35.2	—
+ $\mathcal{L}_{\text{homeo}}$ (ours)	84.3	53.8	31.4



Homeostatic loss lifts symmetry and error alignment across datasets.

Headline Numbers

84.3% CIFAR-10 Top-1, asymmetric hEP <i>up from 60.4% without the homeostatic loss — a +23.9-point jump that lands within just 4.3 points of the fully symmetric network</i>	31.4% ImageNet 32 ² Top-1
	55.2% ImageNet 32 ² Top-5
	14.3% F-MNIST err $ \beta =5$

Takeaway

Equilibrium propagation **does not need symmetric weights**: removing finite-nudge bias with a Cauchy integral and penalizing Jacobian asymmetry lets asymmetric networks reach ImageNet-scale accuracy near the symmetric ceiling — a biologically plausible route toward training physical neural substrates.